

MATERIAL SCIENCE AND METALLURGY

II B.TECH I Semester : MECHANICAL ENGINEERING

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6ME10	PCC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100

COURSE OVERVIEW:

This course is intended to provide basic knowledge on the metallurgical aspects of metals that are being used in our homes to various engineering applications. It covers the fundamental aspects of materials, crystal structures and their representation, and various defects that are present in the materials. Then, the necessity of alloying and the corresponding changes occurring in their phase diagrams are discussed. A special focus is given on the important ferrous and non-ferrous alloys which are extensively used in the industries. Tailoring of material properties through different heat treatment processes with their microstructural changes are included. And, finally the course concludes with discussion on materials other than metals that includes polymers, ceramics and advanced materials such as composites.

COURSE OUTCOMES:

At the end of course, students will be able to

1. Explain basic concepts of crystal structures and their imperfections and also draw crystal points, directions and planes in cubic unit cells.
 2. Interpret various phases present in the binary phase diagrams of alloys and also calculate mass fraction of phases.
 3. Recommend heat treatment processes for the desired changes in properties of steels.
 4. Differentiate ferrous and non-ferrous alloys in terms of their properties and applications.
- Classify and explain polymers, ceramics and composites in terms of their properties and applications.

UNIT-I CRYSTAL STRUCTURE

Metallic crystal structures: Unit cells, crystal systems, directions and planes, slip systems.
Imperfections in solids: point defects, dislocations, surface and volume defects, determination of grain size, strengthening mechanisms for single phase metals: grain size reduction, Hall-Petch equation, cold working.

UNIT-II ALLOYS & PHASE DIAGRAMS

Alloys: Definition, necessity of alloying; Solid solutions: substitutional and interstitial, Hume-Rothery rules. Introduction Phase diagrams: Gibbs phase rule, unary phase diagram, cooling curves of pure metals, construction of phase diagram,
Interpretation of binary phase diagrams: Tie line, Lever rule, various invariant reactions, Microstructure development upon cooling, Iron Carbon phase diagram: Phases, Invariant reactions involved.

UNIT-III HEAT TREATMENT OF STEEL

Transformation Diagrams: Time-Temperature-Transformation (TTT) diagrams Fe-C alloys, bainite and martensite, continuous cooling curves and interpretation of final microstructures and properties.
Heat treatment processes: Annealing & their types, normalizing, spheroidizing, hardening and tempering, Surface hardening: flame and induction hardening, Case hardening: carburizing, nitriding and cyaniding.
 Special heat treatment processes: austempering & its limitations, martempering.

UNIT-IV	FERROUS AND NON - FERROUS ALLOYS
Ferrous alloys: Introduction, Classification, Plain carbon steels: high, medium and low carbon steels, alloying of steels: stainless steels, tool steels; Cast irons: grey, white, malleable and spheroidal cast irons Non - ferrous alloys: Copper alloys - brass and bronze; Aluminium and Aluminium alloys.	
UNIT-V	POLYMERS, CERAMICS AND COMPOSITES
Polymers and Ceramics: Introduction, classification, general characteristics, and potential applications, Advanced Ceramics: MEMS, Optical Fibers, Ceramic Ball bearings. Composites: Introduction, classification, general characteristics, and potential applications. Advanced Composites: Hybrid Composites, Structural Composites, Laminar and Sandwich panels, Nano - composites.	
TextBooks:	
1. W.D.Callister, Materials Science and Engineering-An Introduction, Wiley India, 6 Edition. 2. V.Raghavan, Material Science and Engineering, Prentice Hall of India Private Limited. 3. Sidney H.Avener, Introduction to Physical Metallurgy, Tata McGrawhill education(P)Ltd, 2nd Edition	
ReferenceBooks:	
1. U. C. Jindal, Engineering Materials and Metallurgy, Pearson, 2011. 2. Kenneth G. Budinski and Michael K. Budinski, Engineering Materials, Prentice Hall of India Private Limited, 4th Indian Reprint, 2002. 3. V. D. Kodgire, Material Science and Metallurgy for Engineers, Everest, 1st Edition, 2006. 4. Ashby, "Engineering Materials Vol I & II" fifth edition authored by M F Ashby and David R H Jones.	